

TIM ZHANG

Data Engineer | AI-Driven Pipeline Architect | Data + AI Systems Builder

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Portfolio: portfolio.timzhang.cn | GitHub: github.com/February13

PROFESSIONAL SUMMARY

Data Engineer and AI-driven pipeline architect with 8+ years of production data engineering experience across commercial banking, multinational retail, cloud data warehouses, GovTech automation, insurance FWA, and research AI. Delivered 10TB+ cloud migration, five-layer warehouse architecture, SOX-aware financial data workflows, StarRocks performance optimization, and production reporting pipelines before moving into Agentic RAG, Agent Runtime, GraphDB, and research AI systems. Recent flagship work includes the award-winning SRR Agentic Case Processing System, Uniflo software-hardware multi-end intelligent continuous delivery platform, MCC FWA insurance fraud graph intelligence system, and GaitGPT privacy-first gait research assistant.

TECHNICAL SKILLS

Programming & Data: Python, SQL, SparkSQL, Impala SQL, Bash/Shell, Java

Data Engineering: Hadoop, Spark, Hive, StarRocks, Kafka, Seatunnel, DolphinScheduler, Informatica PowerCenter

AI/LLM Systems: FastAPI, LangChain/LangGraph, RAG, Agentic workflows, Agent Runtime, multi-end delivery, workspace permissions, workflow trace, pgvector, GraphDB/Neo4j, OpenAI GPT-4o, Gemini, LLM-as-Judge

Financial & Retail Data: Commercial-bank credit/limit/lending models, retail PNL/B2B/HR/Leasing reporting, claims adjudication, SOX-aware data workflows

Cloud & DevOps: Google Cloud Run, Cloud SQL, Tencent Cloud, Docker, Linux, GitHub Actions

SELECTED AI & DATA PROJECTS

SRR Agentic Case Processing System | *Technical Lead* | 99.4% contribution | Lingnan Cup 2nd Prize | Featured on TV | 2025-2026

Project links: [Portfolio case study](#) | [Public showcase repo](#)

- Situation/Task: Hong Kong SRR case handling relied on multi-channel materials, manual routing, historical lookup, and repeated A-Q field interpretation.
- Action: Designed a 7-layer agentic architecture with 17 atomic capabilities covering ICC 1823/TMO/RCC parsing, A-Q extraction, department routing, pgvector + RRF retrieval, reply drafting, evaluation, and rollback.
- Result: Delivered 12/12 requirements and 80,430 lines across internal iterations; converted the system into an award-winning case with TV coverage, public reference code, and patent-prep narrative.

Tech: Python, FastAPI, React/TypeScript, PostgreSQL 15 + pgvector, GPT-4o, Google Cloud Run, Cloud SQL, Docker

Uniflo: Multi-End AI Continuous Delivery Platform | *Core Builder* | *Product & Runtime Architecture* | *Software-hardware multi-end delivery* | 2026

Project links: [Portfolio case study](#) | [OpenUniflo repo](#)

- Situation/Task: Fragmented files, meetings, chats, device-facing capture inputs, and tool calls need persistent context, safe access, and repeatable delivery across software and hardware-facing surfaces beyond one-off chat.
- Action: Built across the Python runtime, React Web UI, OpenAI-compatible API, skill system, workspace and capture-aware access boundaries, permission guards, trace events, and Application candidate lifecycle.
- Result: Turned UniFlo from product thesis into an evolving platform codebase and evidence-backed portfolio story, with patent-prep directions around FileKG, Vision-Link Discovery, multi-end capture, and traceable source-reference mechanisms.

Tech: Python, FastAPI/aihttp, React/Vite, OpenAI-compatible API, multi-end workflow orchestration, workspace tools, skill runtime, trace store, permission policy

MCC FWA: Insurance Claims Fraud Graph Intelligence System | *Core Developer* | *Enterprise collaboration* | *InsurTech / Healthcare AI* | 2026

- Situation/Task: Insurance FWA review has sparse fraud labels and relationship-heavy risk signals across patients, providers, diagnoses, receipts, billing items, and amount patterns.
- Action: Converted 38,659 medical claim JSON files into a 1.61M-node / 2.57M-edge property graph, modeled major claim entities, and built portable CSV extraction, Neo4j path validation, and FastAPI async review support.
- Result: Shifted the review surface from opaque approve/reject labels to auditable evidence trails, risk reasons, related entities, policy checks, and claim-level explanations.

Tech: Python, FastAPI, JSON/CSV pipelines, Neo4j, Google Spanner Graph-ready design, LangChain/LangGraph, OpenRouter LLMs

GaitGPT: Privacy-First Clinical Gait Research Assistant | *Architecture Lead* | *Manuscript in preparation* | *iCAN 2026* | *University IP materials in process* | 2026

Project links: [Portfolio case study](#)

- Situation/Task: Gait datasets produce dense sensor indicators, while clinicians reason in symptom terms such as limping, shuffling gait, and asymmetry.
- Action: Led system direction and architecture; implemented intent routing, rule-first CSV analysis, literature-enhanced explanation, structured Findings output, and biomechanics-aware validation.
- Result: Evaluated on a 20-question NONSD-Gait benchmark with 0.9223 overall accuracy and 10.07s median response time; prepared manuscript, iCAN 2026 package, and university IP materials while keeping patent-sensitive implementation details private.

Tech: Python, FastAPI, Next.js/React, SurrealDB, LangGraph, PubMed/Semantic Scholar APIs, Ollama/Qwen, OpenAI-compatible providers

ENTERPRISE DATA ENGINEERING EXPERIENCE

Data Engineer | Shenzhen RenruiHR Technology Co., Ltd. | 09/2023 - 09/2025

- Situation/Task: Retail finance and operations reporting for a multinational enterprise covered PNL, B2B, HR, Leasing, order, contract, billing, inventory, gift-card, warehouse, and finance sources.
- Action: Migrated 10TB+ IDC warehouse workloads to Tencent Cloud, rebuilt ODS/DIM/DWD/DWS/ADS layers, supported B2B refined management, mapped data assets and lineage, and added T+0 data quality response plus production monitoring.
- Result: Improved the finance data foundation by 50%+; Seatunnel sync reached 99.95% consistency and cut failures by 70%; ORC/SparkSQL optimization reduced disk by 38%, improved query speed by 55%, cut ETL runtime by 87.5%, and StarRocks materialized-view / model optimization moved core analytical queries from 3-10 minutes to avg. 0.8s.

Data Engineer | Hangzhou Yatop Information Technology Co., Ltd. | 04/2021 - 08/2023

- Situation/Task: A commercial-bank data middle-platform needed credit-domain maintenance, limit-system upgrade, core lending migration impact analysis, and reusable model layers across 100+ transaction systems.

- Action: Led credit-domain delivery, maintained warehouse domains, integrated sources with Informatica PowerCenter, delivered SIT/UAT reports, optimized 30+ issue points, landed 15 key limit models, and tuned Impala skew, scripts, and UDF repricing-date logic.
- Result: Improved cross-team model reuse by 50%, stabilized credit / limit / lending reporting, and supported bank-grade delivery through lineage, quality rules, controlled changes, and downstream marts.

Data Engineer | Shenzhen Owned Technology Co., Ltd. (Client: Commercial Bank) | 02/2019 - 03/2021

- Situation/Task: Commercial-bank marts needed stable definitions for corporate credit, limits, loan contracts, core lending, loan accounts, loan receipts, disbursement, repayment, settlement accounts, and product-level reporting.
- Action: Maintained warehouse model-layer SQL, translated mart requirements into traceable logic, and aligned source lineage, model changes, issue handling, and reporting definitions with banking stakeholders.
- Result: Supported financial, regulatory, and management reporting with bank-grade delivery discipline: consistent definitions, controlled changes, traceable issue handling, and stable downstream outputs.

Engineer | Chengdu HQtimes Technology Co., Ltd. | 07/2017 - 11/2018

- Situation/Task: IBM application operations required fast issue response and repeatable support routines.
- Action/Result: Troubleshot network/system issues and built Bash/Python automation scripts to reduce repetitive manual operations.

EDUCATION & AIBA COURSEWORK

MSc in Artificial Intelligence and Business Analytics | Lingnan University, Hong Kong | 2025 - 2026

- Relevant coursework: AI foundations, business data management, analytics programming, blockchain, healthcare analytics, data visualization, generative AI programming, and AI-based optimization.

Bachelor of Management | Shenzhen University, China | 2014 - 2017

LANGUAGES

Languages: Cantonese (Native), Mandarin (Native), English (Professional Working Proficiency)